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Deliverable 1: Verification – Force Balance

Question:

Does the total reaction force equal the applied force of 1000 N?

Does the reaction force in the X-direction equal –500 N?

Applied Load (Defined by Components)

$$F_x = +500 \text{ N}$$

$$F_y = -866 \text{ N}$$

$$F_z = 0 \text{ N}$$

Reaction Force Results (From ANSYS)

$$\text{Reaction Force (X)} = -500 \text{ N}$$

$$\text{Reaction Force (Y)} = +866 \text{ N}$$

$$\text{Reaction Force (Z)} \approx 0 \text{ N}$$

$$\text{Total Reaction Force} = 999.98 \text{ N} \approx 1000 \text{ N}$$

Answer

Yes, the total reaction force is approximately equal to the applied force of 1000 N.

The reaction force components are equal in magnitude and opposite in direction to the applied force components. This confirms that force equilibrium is satisfied and the vector load has been applied correctly.

Deliverable 2: Total Deformation

Question:

Since the force has both X and Y components, does the bracket twist or simply bend diagonally?

Total Deformation Results (From ANSYS)

Maximum Total Deformation = 1.7977×10^{-4} m

Minimum Total Deformation = 0 m

The maximum deformation occurs at the free end of the horizontal leg, while the minimum deformation occurs at the fixed support.

Answer

The bracket undergoes combined axial stretching and bending, not twisting.

The X-component of the force causes axial tensile deformation, while the Y-component causes bending of the horizontal leg. Since no force or moment is applied about the longitudinal axis, torsional deformation is not significant.

Deliverable 3: Stress Singularity Check

Question:

Which stress location is real and which is an artifact of the boundary condition?

Location A: Inner Fillet (Curved Corner)

Equivalent (von Mises) Stress Result:

Maximum Equivalent Stress $\approx 7.55 \times 10^7$ Pa

Answer:

The high stress observed at the inner fillet is real stress.

It is caused by actual load transfer and geometric stress concentration at the curved junction of the bracket.

Location B: Fixed Support Sharp Corners

Observation:

Very high stress is observed at the sharp edges of the fixed support.

After mesh refinement to 0.5 mm, the stress value increases by more than 20%.

Answer:

The high stress at the fixed support sharp corners is a stress singularity (fake stress).

It is a numerical artifact caused by the idealized fixed boundary condition and does not represent real material behavior.